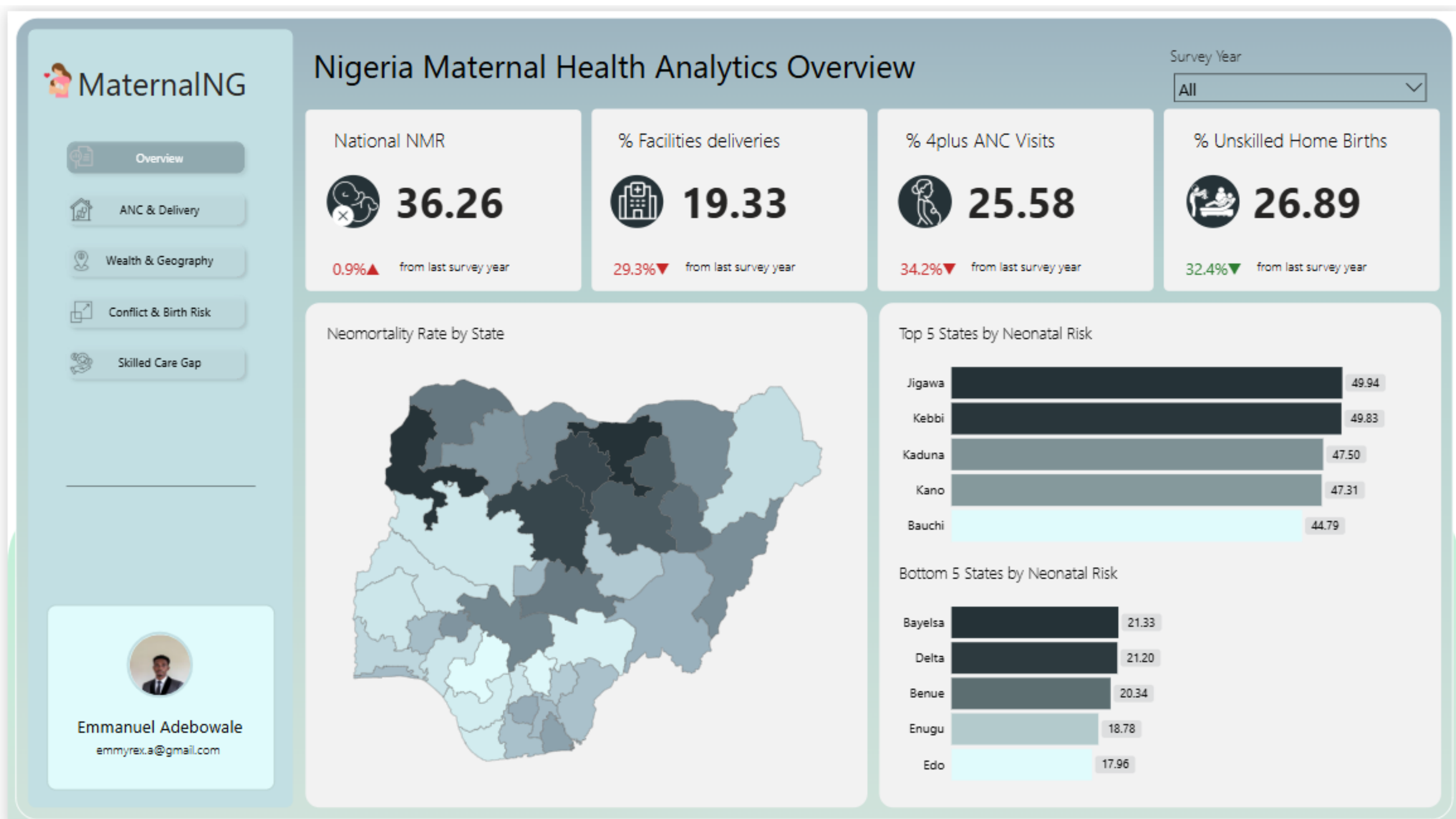
NIGERIA MATERNAL & NEWBORN HEALTH: A DATA PIPELINE ANALYSIS

1: EXECUTIVE SUMMARY

Nigeria accounts for roughly one in five maternal deaths globally. That number has not moved meaningfully in decades, and the data infrastructure to understand why has largely not existed at the state level, where health decisions actually get made.

Figure 1: National overview: neonatal mortality rate, facility delivery coverage, and ANC attendance across Nigeria's 37 states (2018 and 2024 DHS).



This project builds a production-grade data pipeline that ingests, transforms, and serves maternal and newborn health data across Nigeria's 37 states. It draws on two rounds of the Demographic and Health Survey (2018 and 2024), conflict event records from ACLED, and World Bank health expenditure data. The output is a five-page analytical dashboard built in Power BI, backed by a fully orchestrated pipeline running on Apache Airflow, a BigQuery data warehouse modelled in dbt, and a FastAPI serving layer.

Five questions drive the analysis:

1. Does ANC attendance or place of delivery better predict neonatal survival?
2. Does wealth protect against geography for women who delivered without skilled care?
3. Does birth order predict neonatal risk, and where does that risk concentrate?
4. Did conflict intensity reduce ANC attendance in affected states between 2018 and 2024?
5. Does attending ANC with a skilled provider translate to delivering with one?

The findings complicate some common assumptions. Facility delivery does not reliably reduce neonatal mortality as women who arrive at facilities often do so in emergency, which inflates facility NMR. ANC attendance is the stronger predictor. Wealth reduces neonatal mortality within every region, but geography sets the ceiling: a wealthy woman in the North-West faces worse odds than a poor woman in the South-East. And in most states, ANC attendance and skilled delivery are moving in opposite directions.

2: CONTEXT & MOTIVATION

Nigeria has the third highest maternal mortality rate in the world and accounts for nearly 20% of all neonatal deaths globally, as of April 2026. These are not new statistics. What is new is the question of why improvement has stalled and whether the data exists to answer it at the level where it matters.

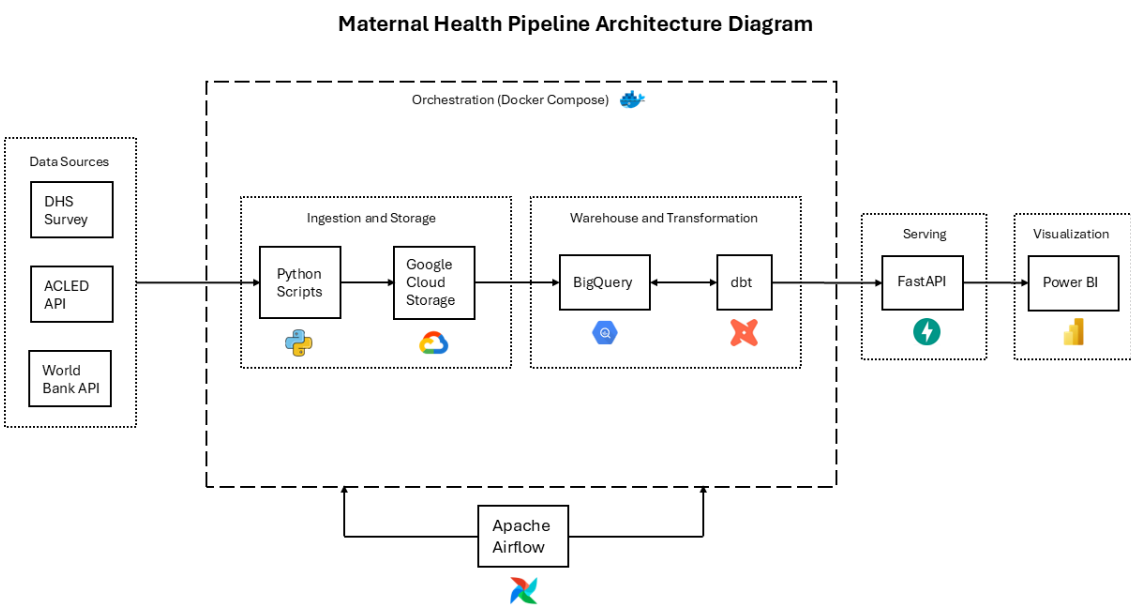
Health policy in Nigeria is implemented at the state level. A national average obscures enormous variation: a state in the South-East and a state in the North-West can sit at opposite ends of nearly every maternal health indicator while sharing the same headline figure. Decisions made without that granularity tend to be blunt instruments.

This project starts from the premise that better analysis requires better infrastructure. The Demographic and Health Survey provides the most reliable subnational data available on maternal and newborn outcomes in Nigeria. ACLED tracks conflict events at the local level. The World Bank publishes health expenditure trends over time. None of these sources were designed to talk to each other, which is exactly the problem this pipeline solves.

3: ARCHITECTURE OVERVIEW

The pipeline moves data through six stages: ingestion, raw storage, warehousing, transformation, serving, and visualization. Apache Airflow orchestrates the first four. The last two operate independently; FastAPI responds to requests on demand, and Power BI connects to it directly.

Figure 2: End-to-end architecture of the Nigeria Maternal Health Pipeline, from raw data ingestion through cloud storage, warehouse transformation, and API serving to the analytical dashboard.



Three data sources feed the pipeline. The Nigeria Demographic and Health Survey (DHS) provides birth and individual recode files from 2018 and 2024 — the most granular subnational data available on maternal and newborn outcomes in Nigeria. ACLED supplies conflict event records at the local government level. The World Bank API provides health expenditure indicators over time. None of these arrive in a compatible format, which is why the ingestion layer exists as custom Python scripts/modules rather than a generic connector.

Raw files land in Google Cloud Storage before anything touches them. This separation matters: it preserves the original data and makes the pipeline recoverable if a transformation breaks downstream.

BigQuery holds the warehouse. dbt Core handles all transformation logic in three layers: staging models that clean and type-cast raw data, intermediate models that join and reshape it across sources, and mart models that produce the aggregated tables the dashboard and API actually query.

FastAPI sits between the warehouse and Power BI. Rather than connecting Power BI directly to BigQuery, the API serves pre-defined analytical endpoints, one per research question. This keeps query logic in one place and makes the analysis reproducible outside the dashboard.

4: FINDINGS BY RESEARCH QUESTION

Q1 — ANC Attendance vs. Place of Delivery

ANC attendance is a stronger predictor of neonatal survival than place of delivery. That finding cuts against a common assumption in maternal health policy, that getting women into facilities is the primary lever for reducing neonatal deaths.

The data tells a more complicated story. Across all ANC categories, facility NMR exceeds home delivery NMR. The data does not explain why, but one plausible mechanism is emergency referral bias. Women arriving at facilities already in crisis inflate the facility death count, while uncomplicated home deliveries go unrecorded. This interpretation is consistent with the pattern but is not directly tested here.

What the data does show clearly is that adequate ANC (four or more visits with a skilled provider) reduces neonatal mortality regardless of where a woman delivers. Women with no ANC who delivered with a skilled attendant recorded the highest NMR in the dataset at 66.95. **Attendance, not location, is the variable that moves outcomes**.

Several states complicate this further. Imo, Bauchi, Jigawa, Kebbi, and Kaduna all show high NMR despite relatively adequate ANC attendance. The data records attendance, not visit content, so whether this reflects variation in care quality, provider skill, or facility resources is a question this dataset cannot answer.

Figure 3: Neonatal mortality rate by place of delivery and ANC adequacy

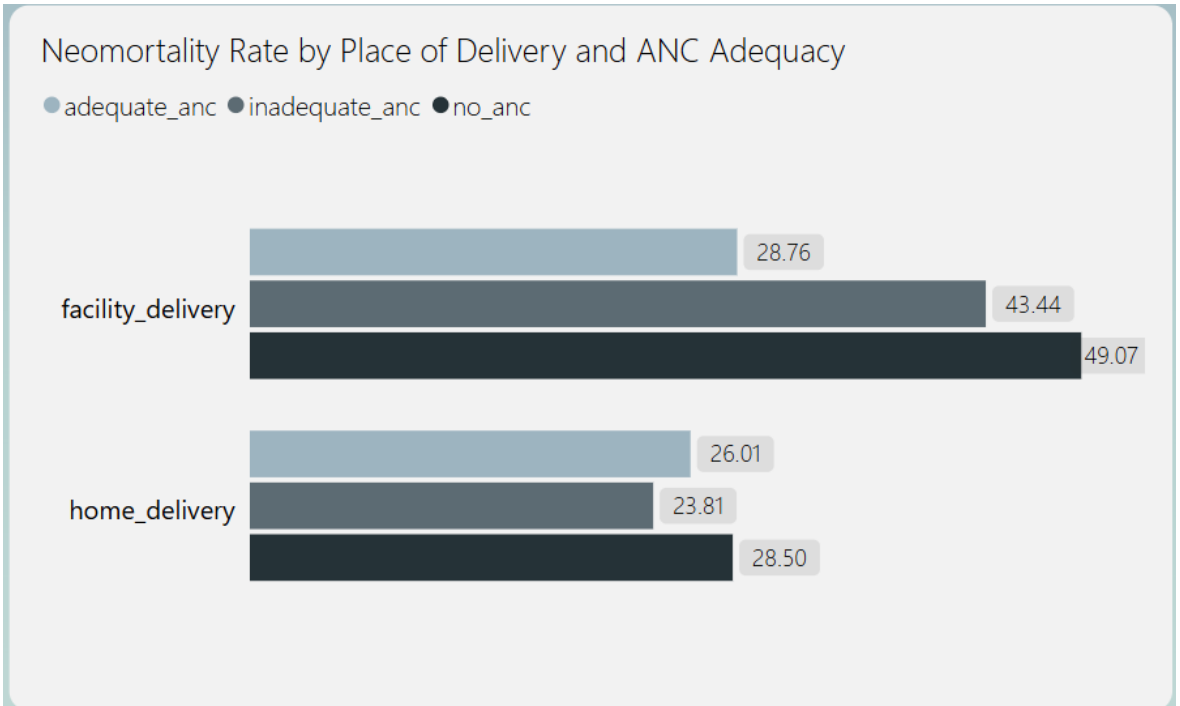
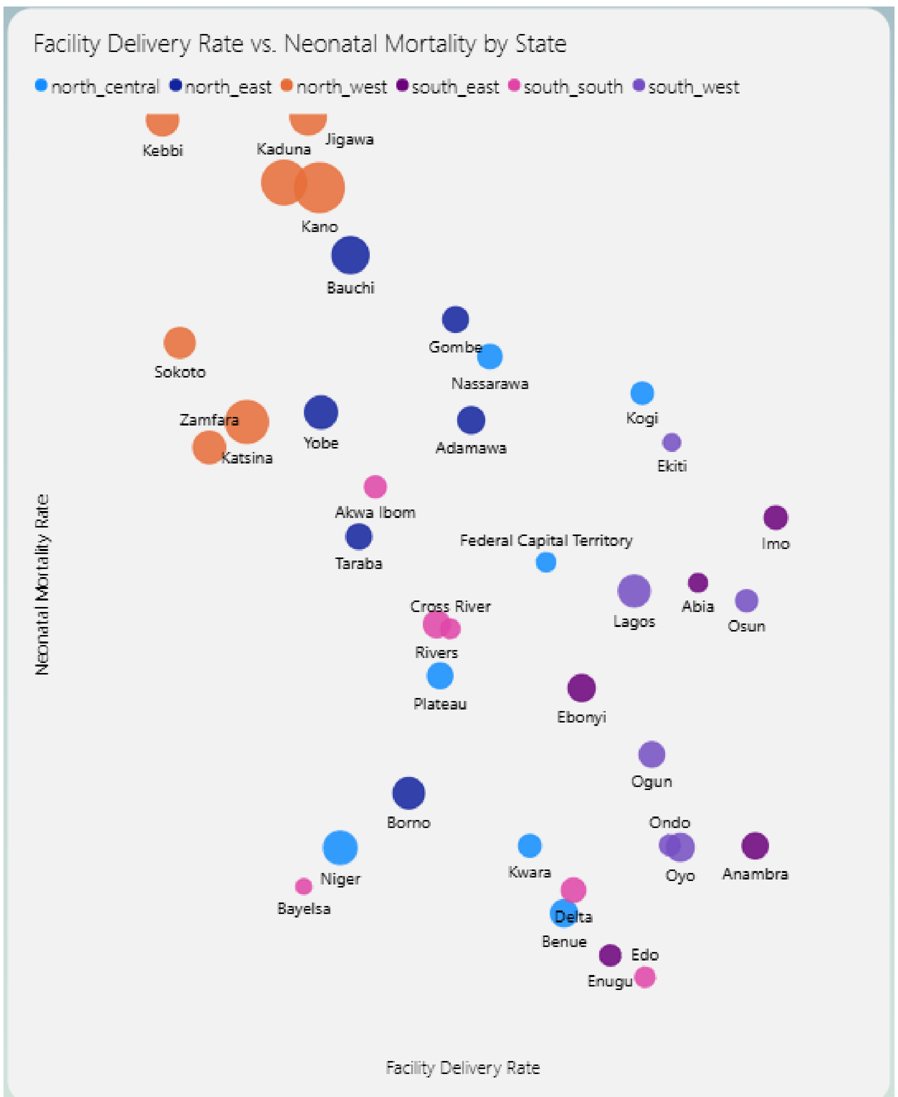


Figure 4: Facility delivery rate versus neonatal mortality rate by state. Zone colour indicates geopolitical region; bubble size represents weighted birth volume.



Q2 — Wealth & Geography

Wealth reduces neonatal mortality, but it does not override geography. Across every geopolitical zone in Nigeria, wealthier women have better outcomes than poorer women in the same region. The gradient is consistent. What changes is the ceiling that geography sets.

A wealthy woman in the North-West faces worse neonatal outcomes than a poor woman in the South-East. Individual economic status improves a woman's odds within her context, but it cannot compensate for what her region lacks: infrastructure, skilled providers, functional referral systems (the chain connecting community care to emergency obstetric services). Zone-level factors dominate where individual factors cannot reach.

The North-West is the starkest case. Wealth provides the least protection there of any zone, suggesting that the gap between rich and poor women narrows not because poorer women are better served, but because even wealthier women face structural barriers that money alone cannot resolve. The data surfaces this pattern but does not identify which structural factors are driving it.

Lagos presents a different anomaly. It has the highest wealth concentration of any state yet sits at a mid-range NMR. One plausible explanation is urban inequality, in which aggregate wealth masks deep disparities within the state, and a high average obscures a wide distribution. Again, this is interpretation; the data shows the outcome, not the mechanism.

Figure 5: Neonatal mortality rate by wealth quintile, 2018 and 2024. Wealthier women record lower NMR within every zone, but the gradient narrows in the North-West.

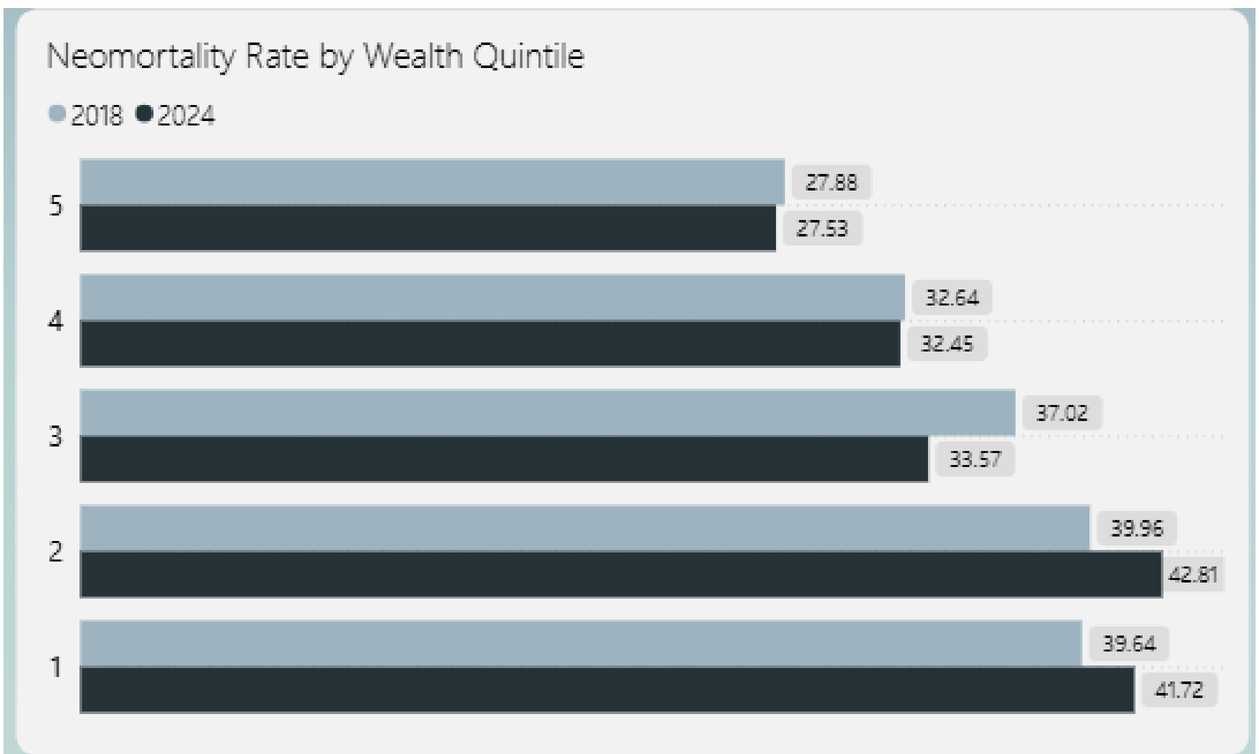


Figure 6: Neonatal mortality rate by geopolitical zone and wealth quintile. Geography sets the ceiling that individual wealth cannot override.

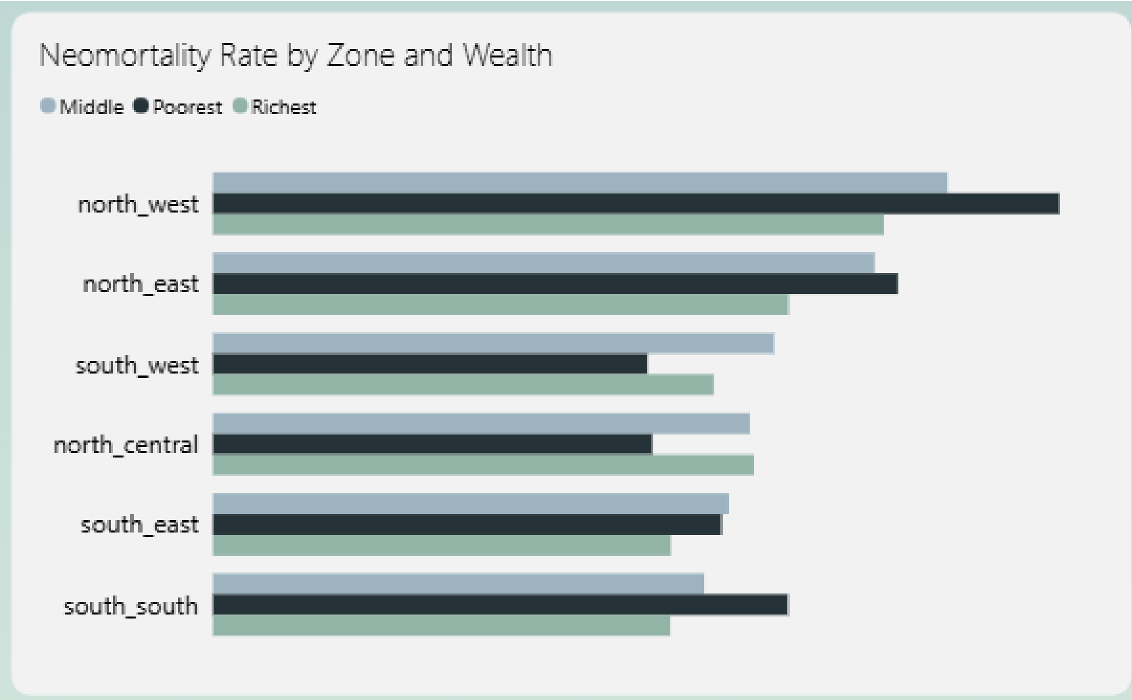
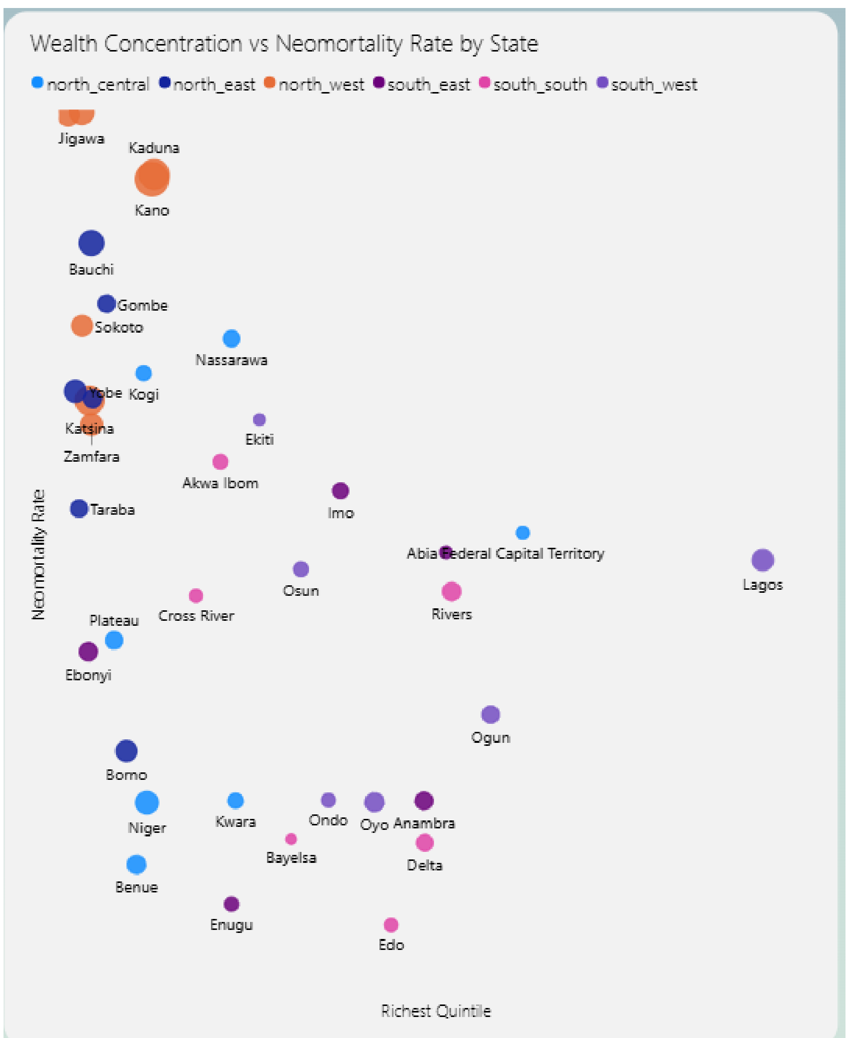


Figure 7: Wealth concentration versus neonatal mortality rate by state. States with higher proportions of wealthy households do not consistently achieve lower NMR.



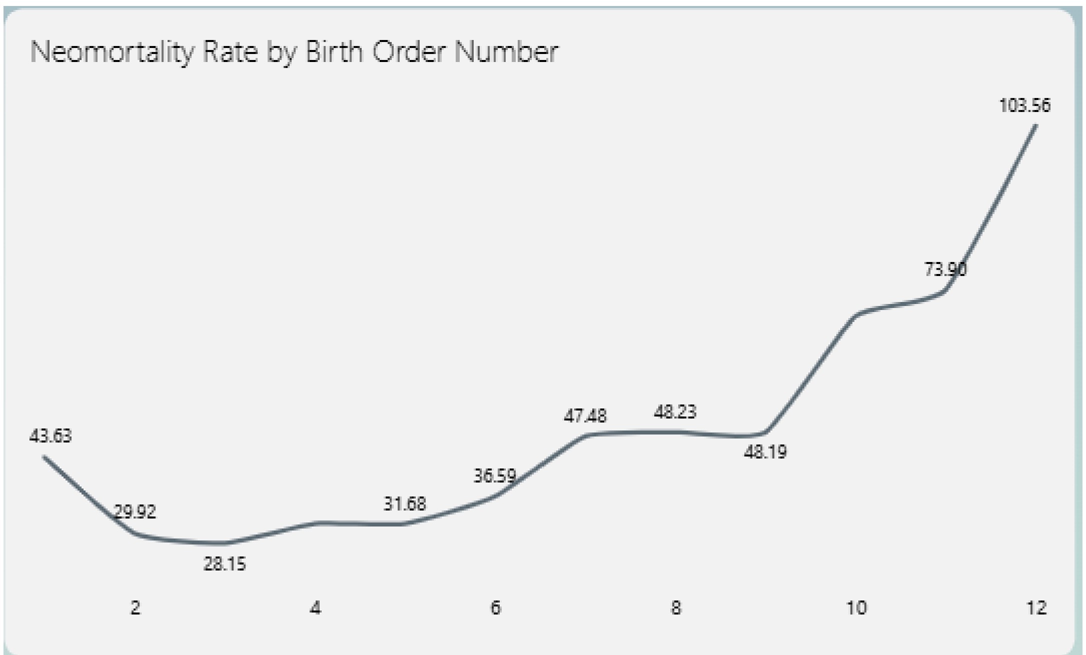
Q3 — Birth Order & Neonatal Risk

First births carry the highest neonatal risk. Risk declines through the second and third birth, then rises again at higher birth orders. The pattern is not linear, but the concentration of risk at the extremes is consistent across states.

The elevated risk at first birth is well documented in maternal health literature: inexperience, physiological factors, and provider readiness all play a role. The uptick at higher birth orders is a separate phenomenon. Maternal depletion (the cumulative effect of repeated pregnancies with insufficient recovery time) is one plausible explanation documented in reproductive health literature, but this dataset does not measure inter-pregnancy intervals or nutritional status directly. The pattern is visible in the data; what drives it here is not.

High birth order records in this dataset carry a reliability caveat. Sample sizes thin out at birth orders above six, making NMR estimates statistically unstable. The analysis flags these as unreliable and excludes them from the visual to avoid overstating a pattern the data cannot confidently support.

Figure 8: Neonatal mortality rate by birth order, stable and reliable records only. Risk peaks at first birth, declines through the third, then rises again at higher orders.



Q4 — Conflict & ANC Attendance

Conflict does not straightforwardly suppress ANC attendance. That is the headline finding, and it complicates a reasonable prior assumption.

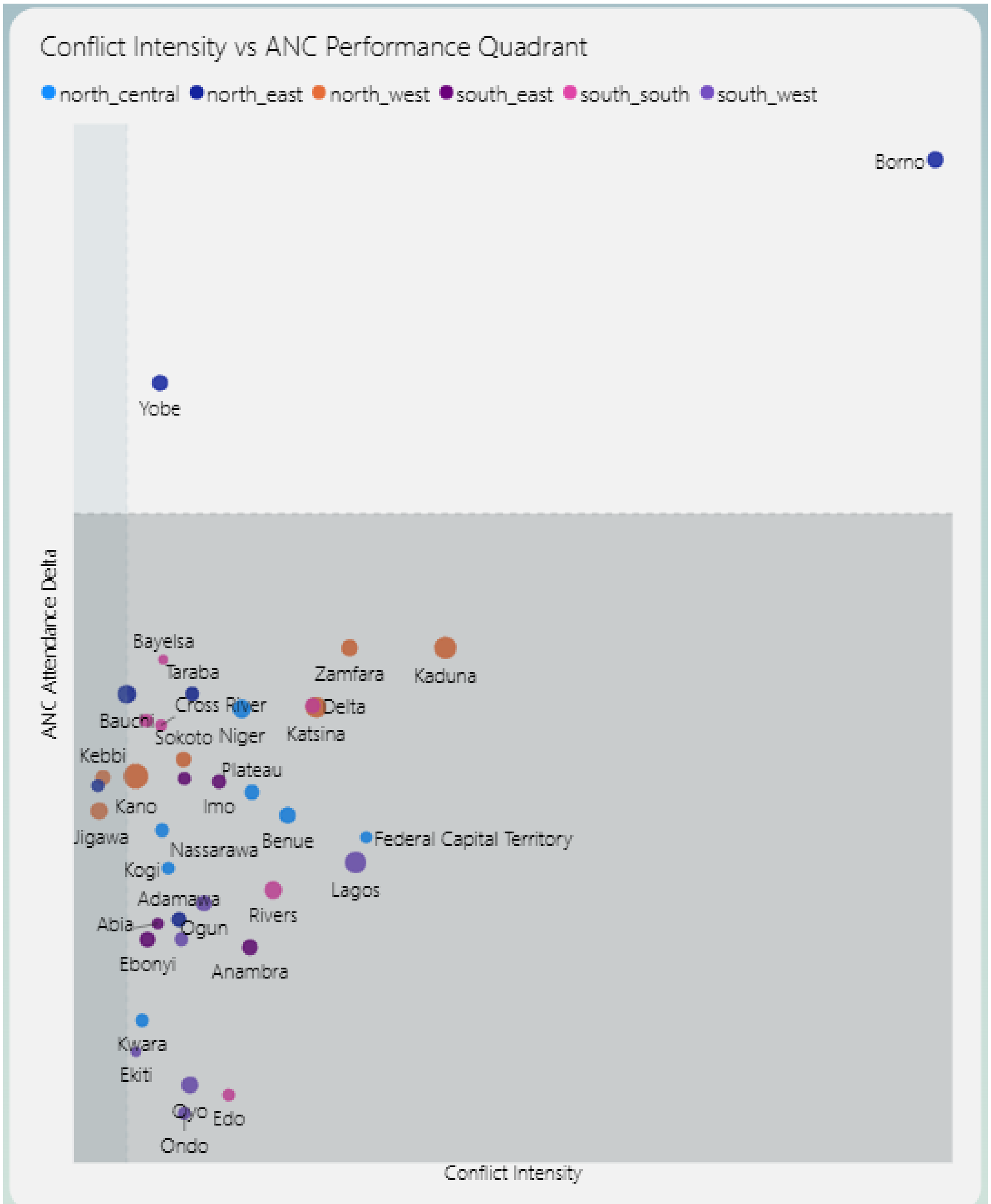
Borno and Yobe, the two states with the highest cumulative conflict intensity in the dataset, both show improvement in ANC attendance between 2018 and 2024. They sit in the top-right quadrant of the analysis: high conflict, positive ANC delta. One plausible explanation is post-conflict intervention: targeted health programmes in high-visibility conflict zones may have partially offset the suppression effect. Displacement patterns are another possibility, where populations moving toward urban centres gain incidental access to care. The data does not distinguish between these mechanisms.

Yobe adds a second layer. When filtered to the 2024 period alone, it shifts leftward, as conflict intensity eased in the second period, and ANC improvement followed. That sequence is consistent with a conflict-suppression effect, but the directionality cannot be confirmed from this data alone.

The more actionable finding sits in the bottom-left quadrant: states with low conflict intensity where ANC attendance still declined. There is no security justification for underperformance in these states. The decline points to system-level failures (funding, provider availability, or demand-side barriers) that are addressable without resolving a conflict first.

Most states declined in ANC attendance regardless of conflict intensity. Conflict is one variable in a larger picture, not the dominant explanation.

Figure 9: Conflict intensity versus ANC attendance change by state, 2018–2024. Bubble size represents weighted respondent volume; quadrant lines sit at median conflict and zero ANC delta.



Q5 — ANC to Skilled Delivery Translation

Attending antenatal care with a skilled provider does not guarantee delivering with one. The gap between the two is where a significant portion of preventable neonatal deaths likely occur, and it varies considerably across states and zones.

The general pattern holds: states with higher ANC attendance tend to have higher skilled delivery rates. Better-functioning health systems perform better on both indicators simultaneously. But the relationship is not uniform, and the exceptions are where the finding becomes actionable.

Several states show ANC attendance improving between 2018 and 2024 while skilled delivery rates did not keep pace. Women are entering the care pathway but not completing it. Whether this reflects cost barriers at the point of delivery, distance to a skilled provider, or a breakdown in the referral chain is not something this dataset can determine. The gap is measurable; the cause requires further investigation.

At the zonal level, the North-West and North-East consistently break the ANC-to-skilled-delivery correlation. High ANC attendance in these zones does not translate to a corresponding skilled delivery rate, suggesting that the barriers to completing care are structural and zone-specific rather than individual.

Figure 10: ANC attendance rate versus skilled birth attendant rate by state. Zone colour indicates geopolitical region; states above the trend line translate ANC gains into skilled delivery more effectively.

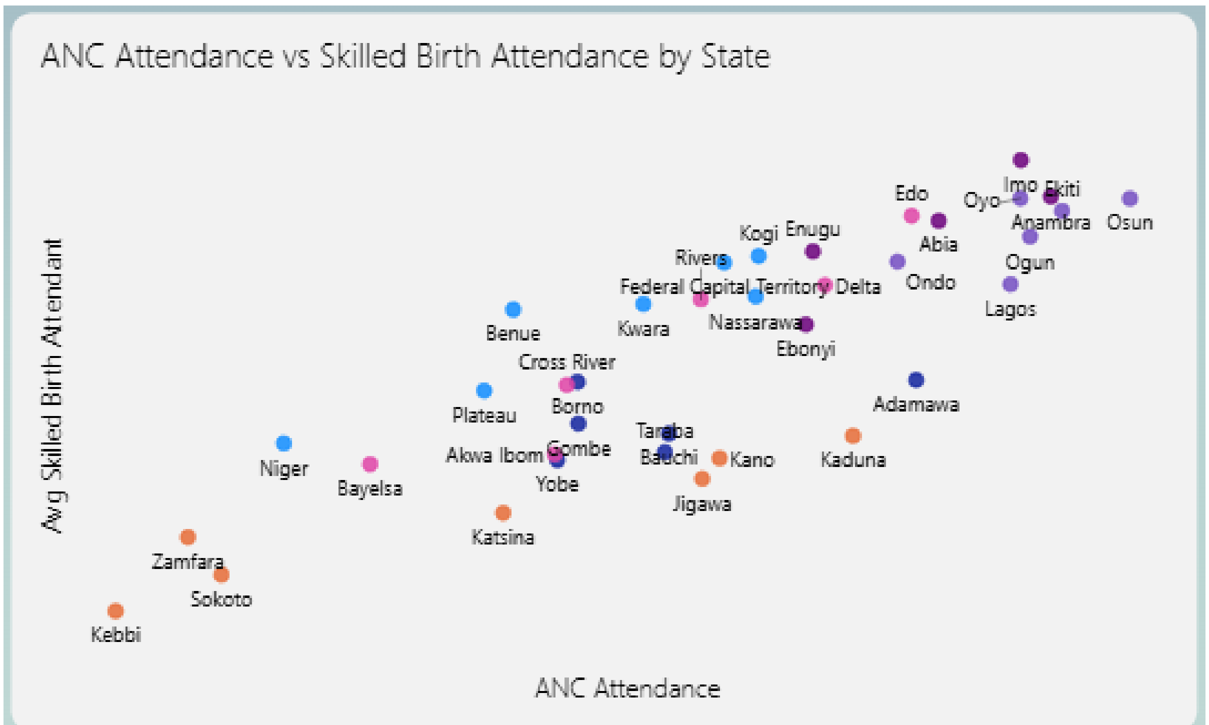


Figure 11: Change in ANC attendance versus change in skilled delivery rate by state, 2018–2024. States where the bars diverge signal a breakdown in the care pathway.

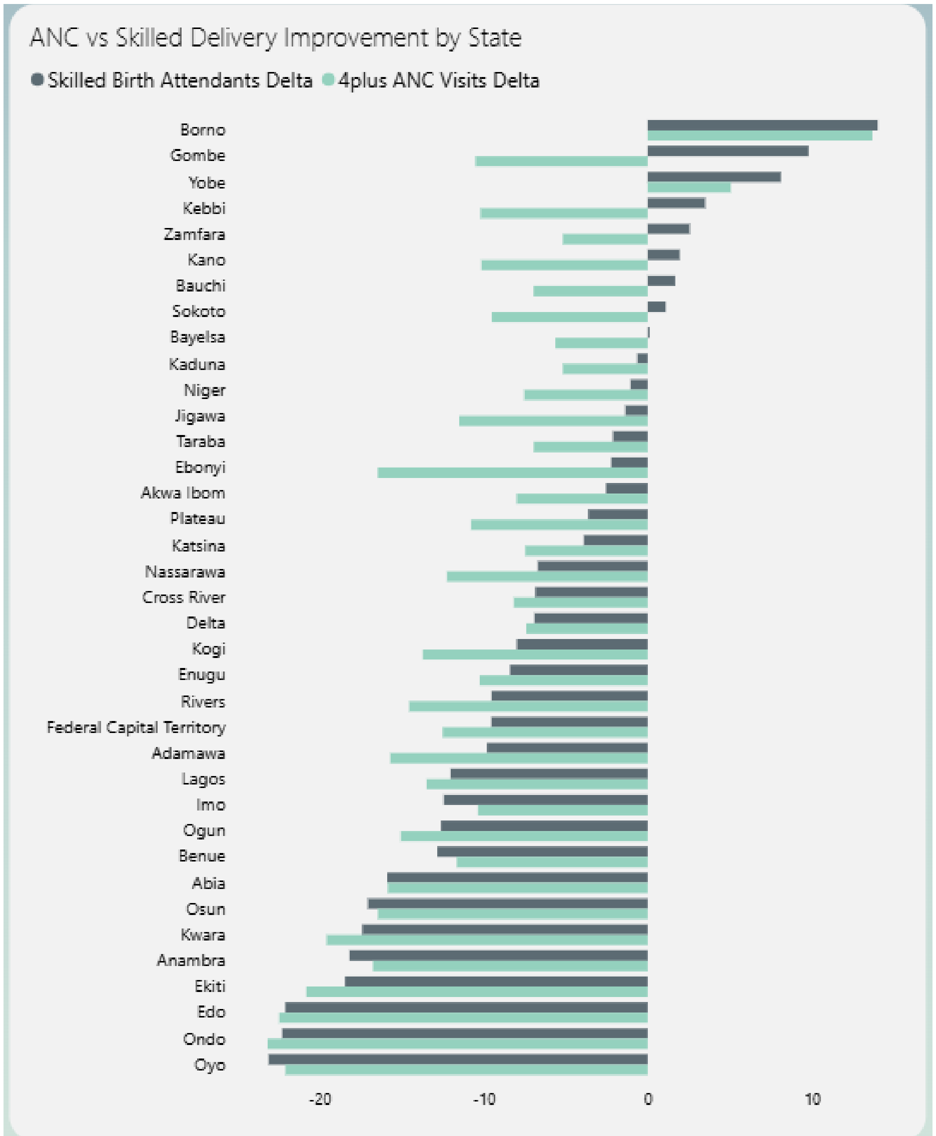
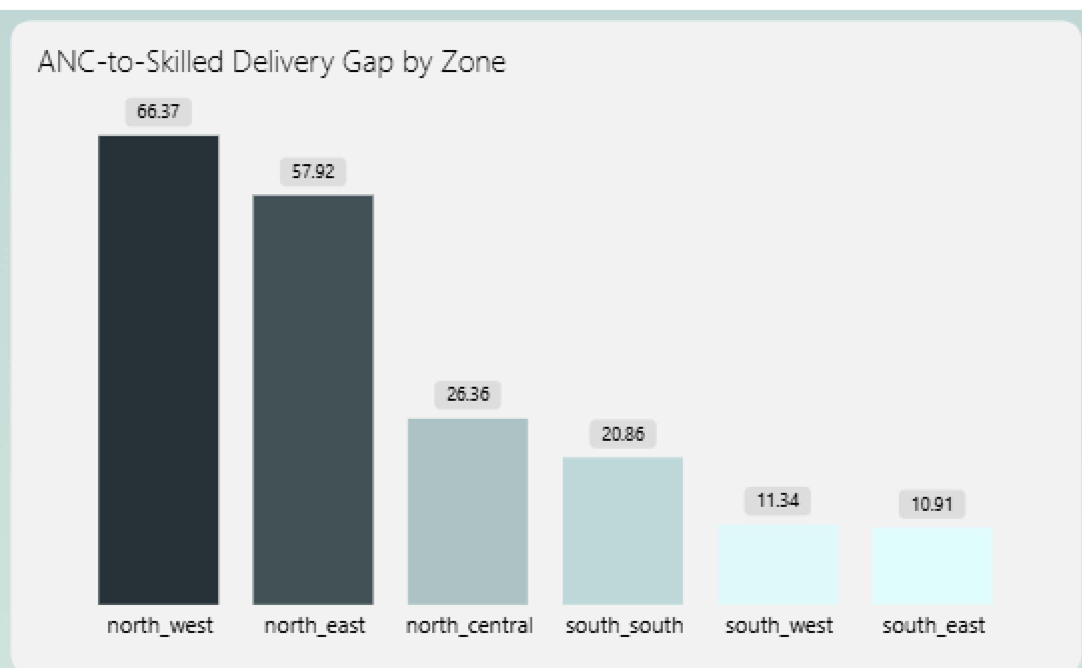


Figure 12: Proportion of women who attended ANC with a skilled provider but did not deliver with one, by geopolitical zone



5: ENGINEERING DECISIONS WORTH NOTING

Several decisions in this pipeline deviate from convention in ways worth explaining.

Aggregate-first data modelling. Classical dimensional modelling (the Kimball approach) builds granular fact tables and lets the BI layer aggregate on the fly. This pipeline inverts that. Mart models are pre-aggregated at the state and survey period level, and the two seed files (dim_state and dim_survey_period) serve as relationship anchors in Power BI rather than full star schema dimensions. For a project driven by five fixed research questions at a known grain, this is a deliberate trade-off: query performance and model clarity over flexibility.

Weighted averages over simple averages. Averaging ANC attendance percentages across states of different population sizes produces a distorted national figure. A state with 200 respondents and a state with 20,000 respondents carry equal weight in a simple average, which is analytically indefensible. All percentage measures in this pipeline use respondent-weighted averages, enforced both in dbt mart models and in DAX measures in Power BI.

DHIS2 exclusion. The District Health Information System was an intended data source. It was excluded after access required Ministry of Health credentials that are not publicly available. This is documented as a finding about data infrastructure in Nigeria, not treated as a project limitation; the inaccessibility of routine health system data is itself analytically relevant.

Reliability flagging for birth order analysis. Sample sizes thin at high birth orders, making NMR estimates statistically unstable. Rather than silently including or excluding these records, the pipeline classifies them explicitly: stable (50 or more records), reliable (25 or more), and unreliable (below 25). The dashboard filters to stable and reliable only, and the classification logic lives in the mart model, not in the BI layer, so it travels with the data regardless of how it is consumed.

FastAPI as a serving layer. Connecting Power BI directly to BigQuery was an option. A FastAPI layer was chosen instead to centralise query logic, version-control the analytical endpoints, and keep the dashboard decoupled from the warehouse. Each endpoint maps to one research question, which means the analysis is reproducible and callable outside Power BI without re-writing any SQL.

6: LIMITATIONS & DATA HONESTY

Every dataset has a boundary, and analysis that does not name its boundaries is not trustworthy.

The DHS survey is conducted every five to six years. The 2018 and 2024 rounds are the two most recent for Nigeria, which means this analysis captures two points in time, not a continuous trend. Changes between those points could reflect genuine shifts in outcomes, changes in survey methodology, or both. The data cannot distinguish between them.

DHS records attendance, not content. A woman counted as having completed four antenatal visits may have received four very different levels of care depending on her provider, her facility, and her state. The pipeline measures coverage; it cannot measure quality. This is the most significant analytical constraint in the findings, particularly for Q1 and Q5.

ACLED conflict data is event-based. It records documented political violence, not the full texture of insecurity: displacement, informal curfews, road closures, or generalised fear that keeps women from seeking care. Cumulative event counts are a reasonable proxy for conflict intensity, but they are a proxy, not a direct measure of how conflict affects health-seeking behaviour.

World Bank health expenditure data is available at the national level only. It appears in the pipeline and in the dashboard but does not contribute meaningfully to state-level findings. A state-disaggregated health expenditure dataset would have allowed direct analysis of spending against outcomes. That data does not exist in a publicly accessible form for Nigeria at the granularity this project required.

DHIS2, Nigeria's routine health information system, was investigated as a data source and excluded when access required Ministry of Health credentials unavailable to independent researchers. Routine facility-level data would have added a third time dimension and allowed longitudinal tracking between survey rounds. Its inaccessibility is a constraint on what any independent analysis of Nigerian maternal health can achieve.

7: RECOMMENDATIONS

These recommendations follow directly from the findings. They are addressed to state-level health planners and policymakers, since that is the level at which the data is most actionable.

Prioritise ANC quality, not just coverage. Attendance rates are recorded; visit content is not. States with adequate ANC coverage but persistently high NMR (Imo, Bauchi, Jigawa, Kebbi, Kaduna) are the clearest signal that counting visits is insufficient. Supervision frameworks that assess what happens inside an antenatal visit are a more useful instrument than attendance targets alone.

Investigate the ANC-to-delivery gap in the North-West and North-East. Women in these zones are entering the care pathway but not completing it. The barrier is not awareness, as ANC attendance exists. Something breaks between the last antenatal visit and the moment of delivery. Qualitative research at the facility and community level in these zones would identify whether the gap is financial, geographic, or structural.

Address low-conflict, low-ANC states as a distinct policy problem. States in the bottom-left quadrant (low conflict intensity, declining ANC attendance) have no security justification for underperformance. These states warrant direct programmatic attention, separate from the conflict-affected response that dominates federal maternal health resourcing.

Invest in data infrastructure. The inaccessibility of DHIS2 to independent researchers means that routine health system data like facility utilization, provider availability, stock-outs, cannot be incorporated into any analysis conducted outside the Ministry of Health. Expanding access to anonymized routine data would significantly improve the quality of evidence available for state-level decision-making.

Disaggregate health expenditure reporting to the state level. National health expenditure figures mask the same variation that national NMR figures do. Without state-level spending data, it is impossible to assess whether resources are reaching the states where outcomes are worst.